

Single Source Justification (Simplified Acquisition)

Section A: General Contract Information

Is this a Bridge Action as defined in the [AF Bridge Action Reduction Plan](#)? ☐ Yes ☒ No

Contracting Activity: 10 CONS / PKB

Project/Program Name: FY25 Crystal X-Ray Upgrade

Purchase Request / Local ID Number: F4BWB05072A001

Estimated Contract Cost (including Options): \$ [REDACTED]

Section B: Description of the Supplies/Services Required

MODIFICATION OF EXISTING SYSTEM

- De-Install current detector(HyPix Bantam) from Synergy-i (S/N: PL19340009) - Detector will be set aside, removal is not included.
- Install HyPix-6000 on Synergy-i (S/N: PL19340009)HyPix-6000HE Hybrid Photon Counting Detector, to include modification / upgrading of the existing Synergy-i system.

Section C: Justification for Soliciting from a Single Source or Brand Name

An upgraded detector is essential for improving USAFA's current capability with a single-crystal x-ray diffraction facility. The current detector is an older unit and has degraded in its performance. The issues with the unit without an efficient upgrade will be: continuously lower data quality due to several dead pixels, increased time to collect high and low resolution data separately, and increased background noise.

The HyPix-6000 upgrade for the Synergy-i is unique to Rigaku Instruments and has components relating to efficiently upgrade the Rigaku Synergy-i. The cost to purchase a new Single Crystal X-Ray Analytical system would generally range from \$50,000 to \$500,000, depending on the sophistication, application, and features of the equipment. According to Subject Matter Experts(SME), High-end models with advanced capabilities and specialized functions are at the higher end of the price spectrum, and could range higher than \$500,000.

Section D: Efforts to Obtain Competition

Internet searches and discussions with our SME regarding the upgrade of this particular equipment is necessary to single source award.

Section E: Steps to Preclude Future Single Source or Brand Name Awards

Due to aging equipment and technological advances in the industry, future single source or brand name determinations could be precluded, should a new system be necessary. In this case, the equipment would be purchased under competitive procedures to replace the existing equipment.

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Section F: Contracting Officer's Determination

I have determined, in accordance with [FAR 13.106-1\(b\)\(1\)](#), that the circumstances of this contract action deem only a single source is reasonably available.

Date	Typed Name and Rank/Grade of Contracting Officer	Signature of Contracting Officer
29-Apr-2025	KRISTIN A. HEIKKILA, NH-03, Contracting Officer	 Digitally signed by HEIKKILA.KRISTIN.A.1154229961 Date: 2025.04.29 10:31:57 -06'00'